

Type of the Paper (Article, Review, Communication, etc.)

Predicting Cardiovascular Disease Risk Using XGBoost and LightGBM on Clinical Patient Data

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Abstract

Cardiovascular disease (CVD) is the leading cause of mortality globally, with over 17.9 million deaths annually. Early detection is crucial to reduce morbidity and improve patient outcomes, yet traditional diagnostic methods often fail to identify high-risk individuals in time. This study aims to develop and evaluate machine learning models, specifically XGBoost (XGB) and LightGBM (LGB), to predict the presence or absence of CVD using clinical patient data. Various health metrics, including demographic information, vital signs, and laboratory results, were used as predictors. Model performance was assessed using metrics such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC). Both XGB and LGB demonstrated high predictive performance, with XGB slightly outperforming LGB in most metrics. Feature importance analysis highlighted key risk factors, offering insights into the most influential clinical indicators of CVD. The findings suggest that these machine learning models can serve as effective tools for early detection and risk stratification in cardiovascular healthcare.

Keywords: cardiovascular disease; xgboost; lightgbm; machine learning; risk prediction; clinical data

1. Introduction

Cardiovascular disease (CVD) remains the leading cause of death worldwide, representing a significant public health challenge. According to the World Health Organization (WHO, 2023), CVD is responsible for over 17.9 million deaths annually, accounting for nearly one-third of all global deaths. The high mortality rate is largely attributed to late diagnosis and limited early detection methods, which often prevent timely intervention and effective treatment.

Early and accurate prediction of CVD risk is therefore critical to reduce morbidity and mortality. Recent advances in machine learning offer promising approaches for disease prediction by leveraging complex clinical and demographic data. Among these, eXtreme Gradient Boosting (XGBoost) and Light Gradient Boosting Machine (LightGBM) have shown remarkable performance in various healthcare applications due to their ability to handle large datasets, manage missing values, and capture nonlinear relationships.

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The aim of this study is to develop and evaluate machine learning models, specifically XGBoost and LightGBM, to accurately predict the presence or absence of cardiovascular disease based on various patient health metrics. This research seeks to enhance early detection, identify the most influential risk factors, and contribute to the development of data-driven decision support systems in cardiovascular healthcare. By integrating robust predictive modeling with clinical data, this study aims to support clinicians in making informed decisions and ultimately improve patient outcomes.

2. Theoretical Framework

It consists of concepts and, together with their definitions and reference to relevant scholarly literature, existing theory that is used for your particular study. The theoretical framework must demonstrate an understanding of theories and concepts that are relevant to the topic of your research paper and that relate to the broader areas of knowledge being considered.

3. Materials and Methods

The Materials and Methods should be described with sufficient details to allow others to replicate and build on the published results. Please note that the publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

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4. Results

This section may be divided by subheadings. It should provide a concise and precise description of the experimental results, their interpretation, as well as the experimental conclusions that can be drawn.

4.1. Subsection

4.1.1. Subsubsection

Bulleted lists look like this:

- First bullet;
- Second bullet;
- Third bullet.

Numbered lists can be added as follows:

1. First item;

2. Second item;

3. Third item.

The text continues here.
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All figures and tables should be cited in the main text as Figure 1, Table 1, etc.

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Figure 1. This is a figure. Schemes follow the same formatting.

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Table 1. This is a table. Tables should be placed in the main text near to the first time they are cited.

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Title 1	Title 2	Title 3
entry 1	data	data
entry 2	data	data ¹

¹ Tables may have a footer.

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The text continues here (Figure 2 and Table 2).

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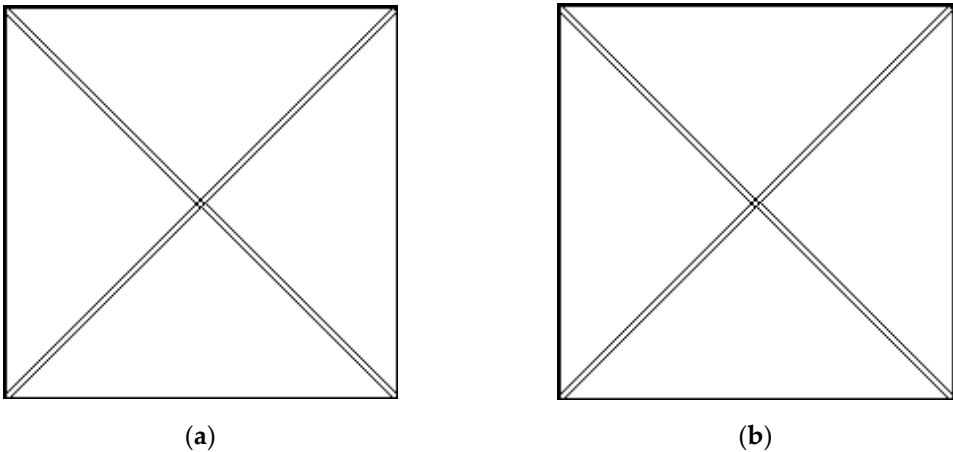


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Table 2. This is a table. Tables should be placed in the main text near to the first time they are cited.

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Title 1	Title 2	Title 3	Title 4
entry 1 *	data	data	data
	data	data	data
	data	data	data
entry 2	data	data	data

	data	data	data
entry 3	data	data	data
	data	data	data
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	data	data	data
entry 4	data	data	data
	data	data	data

* Tables may have a footer.

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4.3. Formatting of Mathematical Components

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This is example 1 of an equation:

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$$a = 1,$$

(1)

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

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This is example 2 of an equation:

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$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

(2)

the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

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Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

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Theorem 1. *Example text of a theorem. Theorems, propositions, lemmas, etc. should be numbered sequentially (i.e., Proposition 2 follows Theorem 1). Examples or Remarks use the same formatting, but should be numbered separately, so a document may contain Theorem 1, Remark 1 and Example 1.*

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Proof of Theorem 1. Text of the proof. Note that the phrase “of Theorem 1” is optional if it is clear which theorem is being referred to. Always finish a proof with the following symbol. □

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The text continues here.

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5. Discussion

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Authors should discuss the results and how they can be interpreted from the perspective of previous studies and of the working hypotheses. The findings and their implications should be discussed in the broadest context possible. Future research directions may also be highlighted.

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6. Conclusions

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This section is not mandatory but can be added to the manuscript if the discussion is unusually long or complex.

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7. Patents

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This section is not mandatory but may be added if there are patents resulting from the work reported in this manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data is shown in the main text can be added here if brief, or as Supplementary data. Mathematical proofs of results not central to the paper can be added as an appendix.

Table A1. This is a table caption.

Title 1	Title 2	Title 3
entry 1	data	data
entry 2	data	data
entry 3	data	data

Appendix B

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Notes

- 1 This is a note example.
- 2 This is a note example.
- 3 This is a note example.

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